

Task value inside the AI time-horizon frontier

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Abstract

METR evaluates AI agents by the maximum length of tasks they can reliably complete. This paper translates that laboratory benchmark into an economic aggregate. By mapping human duration estimates from the Anthropic Economic Index (AEI) to 18,796 O*NET work tasks, I calculate the share of the US wage bill comprising tasks that fall inside the AI capability frontier. Rather than applying a binary threshold, tasks are weighted by the success probability implied by METR’s fitted logistic curves. Under the latest model measured within METR’s reliable range, tasks carrying 60 per cent of the wage bill are expected to succeed. A compound bootstrap accounting for suite composition, duration measurement noise, and imputation yields a 95 per cent interval of 52 to 66 per cent. By contrast, tasks actively delegated to Claude by users account for only 44 per cent of the wage bill cumulatively. At the occupational level, no single occupation has 90 per cent of its task value situated inside the capability frontier. The results indicate that while AI capability encompasses a majority of the wage bill in expectation, it covers parts of most jobs rather than the entirety of a few.

1 Introduction

METR measures the performance of AI agents by the maximum length of the tasks they can autonomously complete, expressed as the time a human would require for the same work (Kwa et al., 2025). On METR’s evaluation suite, this time horizon has doubled roughly every 4.2 months since 2023 (METR, 2026). Concurrently, the Anthropic Economic Index (AEI) tracks which work tasks users delegate to Claude, estimating how long those delegated chunks of work would take a person without AI (Handa et al., 2025; Anthropic, 2026). This paper places both metrics on a single scale—US work tasks weighted by their corresponding wage bill—to measure two distinct quantities: the share of the US wage bill representing tasks a frontier agent is expected to complete, and the share representing tasks workers actually delegate to an AI system in practice.

Macroeconomic scenario models of AI and the labour market, such as the framework developed at Anthropic by Korinek et al. (2026), rely on assumed values for both of these quantities to forecast economic trajectories. To generate their “extreme” 2030 scenario, for example, their model assumes an affected task mass (m_t) covering 50 per cent of all tasks, combined with an automation share (ψ_t) of 0.90, meaning nine in ten AI-performed tasks require no human supervision. Depending on the values chosen for these theoretical dials, the framework produces massive variation in macroeconomic outcomes, ranging from negligible disruption to a scenario featuring 17.9 per cent cognitive

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unemployment, 15.4 per cent annual GDP growth, and the labor share of income plummeting to 45.2 per cent. Establishing an empirical baseline for these parameters is therefore critical.

To evaluate capability, I integrate METR’s fitted logistic curves, which map an agent’s completion probability to the logarithm of a task’s human time. Rather than relying on discrete thresholds—such as the 50 or 80 per cent horizons—I calculate the probability of success for each task given its length, and sum the wage bill weighted by that probability. This expected-success share provides a continuous measure of theoretical capability.

I anchor the estimates using two models: Claude Opus 4.6, the most capable model falling inside METR’s 16-hour reliability range; and Mythos Preview (early), a pre-release checkpoint representing the latest measured frontier. Under Claude Opus 4.6, the expected-success share is 52 per cent of the US wage bill; under Mythos Preview (early) it rises to 60 per cent. Conversely, observed extensive-margin usage indicates that tasks delegated to Claude represent 25 per cent of the wage bill in May 2026, and 44 per cent cumulatively across all AEI releases. Capability and observed usage do not cleanly nest: workers frequently use AI to augment pieces of longer tasks that the agent cannot complete autonomously, while many short, highly-capable tasks remain entirely untouched by users.

This paper builds on task-based models of technical change (Autor et al., 2003; Acemoglu and Restrepo, 2018) and the subsequent literature measuring occupational AI exposure (Brynjolfsson et al., 2018; Webb, 2020; Felten et al., 2021; Eloundou et al., 2023). However, it replaces the subjective expert or model-based rubrics prevalent in that literature with tested laboratory capabilities and logged usage data. Translating these benchmarks to the broader economy requires explicit structural extrapolation: AEI’s duration estimates for delegated chat tasks serve as a proxy for full employer-defined task lengths, and success curves estimated on software engineering tasks are applied across all non-physical work. The empirical strategy embraces this uncertainty, bounding the estimates through a compound bootstrap and a suite of robustness checks.

Section 2 describes the data and task classification rules. Section 3 details the capability measures and out-of-domain assumptions. Section 4 presents the results, and Section 5 details occupational aggregation and structural limitations.

2 Data

2.1 METR time horizons

Figure 1 reproduces METR’s headline capability estimates. Each point represents a model’s 50 per cent time horizon. METR explicitly flags measurements above 16 hours as unreliable due to a scarcity of long-duration tasks in their current suite. Mythos Preview (early) is the only frontier model positioned above this threshold, while Claude Opus 4.6 is the last model reliably measured below it.

METR’s evaluation suite is highly specialized, covering 228 computer-based tasks across 79 families (Appendix Table 7). The majority derive from HCAST—a repository of software engineering, machine learning, cybersecurity, and research tasks with completion times benchmarked on human contractors (Rein et al., 2025). To construct the expected-success shares, I rebuilt every run from the raw per-task success counts and refit the logistic curves using METR’s exact methodology. This refit reproduces METR’s published horizons to within 0.02 per cent for every model (Appendix

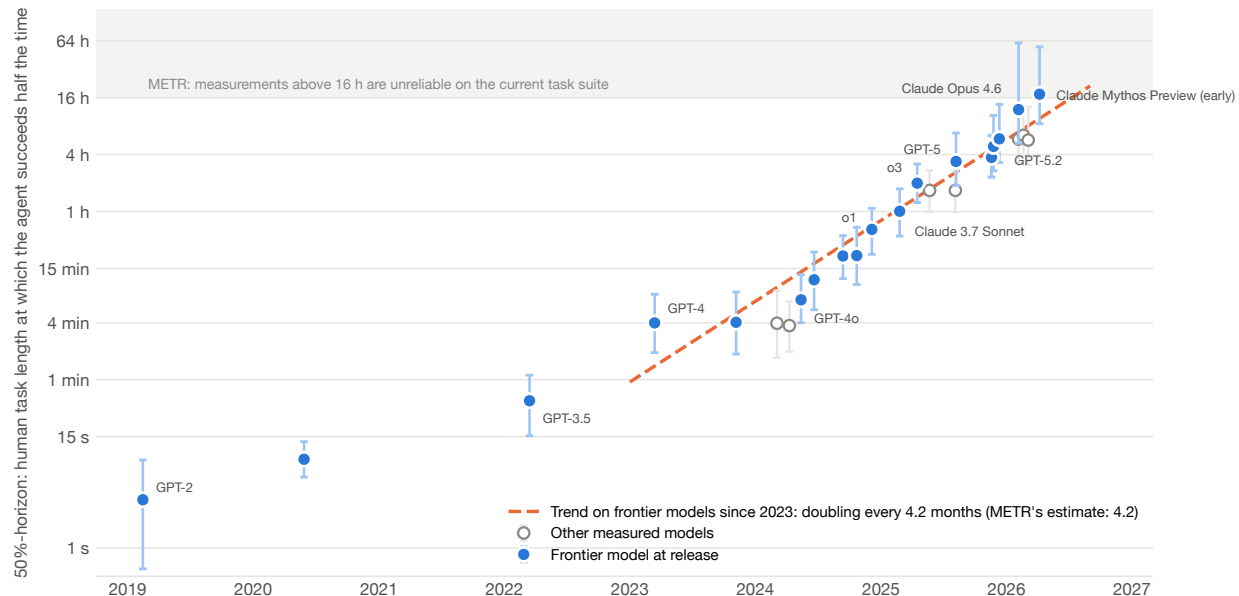


Figure 1: METR’s 50 per cent time horizons, reproduced from METR’s published estimates (METR, 2026). Bars are METR’s published 95 per cent confidence intervals. Filled points: frontier model at release. Dashed line: exponential trend fitted to frontier models released from January 2023 with horizons below 16 hours. Grey band: horizons METR describes as unreliable on the current suite.

Table 8), supplying the coefficients for the continuous success curve.

2.2 The Anthropic Economic Index

The AEI reports the share of Claude conversations mapped to specific O*NET tasks, alongside an AI-generated estimate of the human time required to complete the delegated work without AI (Handa et al., 2025; Anthropic, 2026). For task durations, I rely on the June 2026 release because it maps tasks by their exact O*NET IDs; earlier releases are retained for robustness checks.

Durations are directly observed for 2,814 tasks, representing 26 per cent of the wage bill. For the remaining tasks, I impute the duration using the geometric mean of observed durations among tasks sharing an O*NET detailed work activity (see Appendix A for fallback levels and hold-out validation). The resulting wage-weighted median task duration across the economy is 3.1 hours.

2.3 Wage weights from O*NET and OEWS

I map each O*NET task to its parent occupation and assign it a proportional share of the occupation’s total wage bill based on OEWS May 2025 employment and mean annual wages (U.S. Bureau of Labor Statistics, 2026; National Center for O*NET Development, 2026). Summed over all 18,796 tasks, this yields \$10.72 trillion across 153.4 million jobs. Replacing O*NET importance ratings with an hours-implied weight or adjusting for OEWS annualisation conventions barely moves the aggregate, yielding an expected-success share of 60 per cent (Appendix F).

2.4 Tasks, work activities and the physical filter

The baseline capability estimates employ a deliberately strict physical filter to isolate the capabilities of unembodied software agents. A task is classified as physical if *any* of its underlying O*NET detailed work activities map to a physical generalized work activity (e.g., handling objects, controlling machines, operating vehicles, or assisting and caring for others). It is classified as non-physical only if *none* do. Table 1 illustrates this mapping.

This strict rule enforces the premise that the expected-success share measures the autonomous completion of a task exactly as defined. A software agent cannot complete a mixed task unless an employer unbundles it and reassigns the physical component. Of the 18,796 tasks in the economy, 4,819 are entirely physical and 657 mix physical and non-physical activities. Together, these carry 22 per cent of the wage bill and are assigned a 0 per cent expected success probability.

Table 1: From task to work activity: the O*NET mapping for the highest-importance task in each class for Registered Nurses, and for the highest-importance physical task of Construction Laborers. A task is physical if any detailed work activity it maps to sits under a physical generalized work activity.

Task (occupation)	Detailed work activities → generalized work activity	Counted as
Record patients’ medical information and vital signs. (Registered Nurses)	Record patient medical histories. → Documenting/Recording Information	non-physical
Administer medications to patients and monitor patients for reactions or side effects. (Registered Nurses)	Monitor patient conditions during treatments, procedures, or activities. → Monitoring Processes, Materials, or Surroundings Administer non-intravenous medications. → Assisting and Caring for Others (physical)	physical
Provide health care, first aid, immunizations, or assistance in convalescence or rehabilitation in locations such as schools, hospitals, or industry. (Registered Nurses)	Immunize patients. → Assisting and Caring for Others (physical) Treat acute illnesses, infections, or injuries. → Assisting and Caring for Others (physical)	physical
Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials, such as asphalt. (Construction Laborers)	Operate pumps or compressors. → Controlling Machines and Processes (physical)	physical

Table 2: Task universes. Shares are of the \$10.72 trillion wage bill attached to all tasks.

Universe	Tasks	Share of wage bill
All O*NET tasks in occupations with OEWS employment and wages	18,796	100.0%
Non-physical tasks (no physical work activity)	13,320	78.3%
Tasks with an observed AEI duration (June 2026 AEI release, Apr and May 2026)	2,814	25.6%
Computer and Mathematical occupations (in-domain check, SOC 15)	770	5.9%

The choice of exclusion rule is a first-order determinant of economic exposure (Table 3). Implementing a lenient filter—excluding tasks only if *every* activity is physical—raises the expected-success share under Mythos Preview (early) to 63 per cent. Conversely, applying the [Eloundou et al. \(2023\)](#) exposure labels drops the share to 49 per cent. Defining the boundary of embodied work alters the capability baseline more significantly than a model generation.

Table 3: Three rules for the excluded task set, Mythos Preview (early) as the frontier. Denominator: all tasks. The excluded share is the part of the wage bill that can never count toward the capability numerator.

Rule for the excluded set	Excluded share of wage bill	Expected success	< p80
Strict physical filter (baseline): any work activity physical	22%	60%	31%
Lenient physical filter: every work activity physical	18%	63%	34%
Eloundou et al. (2023) labels: exposure score $\beta = 0$	35%	49%	24%

3 Measures

Let D_t denote task t ’s duration in human hours, and $P_d(D)$ the fitted success curve of the frontier model at date d . The expected-success share is defined as:

$$S_d = \frac{\sum_t V_t P_d(D_t) \mathbf{1}[t \text{ non-physical}]}{\sum_t V_t}.$$

The threshold shares, F_d^{50} and F_d^{80} , replace the continuous probability $P_d(D_t)$ with an indicator taking the value of one if D_t falls below the 50 or 80 per cent horizon. Because S_d integrates the full curve, it captures partial success probabilities for tasks extending beyond the threshold, and proportionally discounts shorter tasks by their residual failure rate. It is strictly a measure of theoretical capability; it does not account for human verification costs or the institutional friction of reassigning standing responsibilities to an agent.

Evaluating S_d across the entire economy requires extrapolating METR’s success curves out of domain. The core assumption is that at a given human duration, an AI agent’s probability of completing a paralegal’s non-physical task equals the probability predicted for a software engineering task of identical length. Internal diagnostics from METR’s suite suggest this is a plausible, albeit imperfect, anchor.

Task length robustly predicts average success within the evaluation suite. For individual frontier models, the weighted R^2 of task success rates on the pooled curve is 0.24 for Mythos Preview (early) and 0.26 for Claude Opus 4.6. Due to binomial noise from small sample sizes (four to six runs per task), a perfectly specified curve would hit an R^2 ceiling of roughly 0.57 on these data, meaning length accounts for approximately 43 per cent of the explainable variation (Appendix D). Because the aggregate share S_d is linear in P_d , it only requires the curve to accurately capture the conditional mean of success at a given length, rather than perfectly predicting the outcome of each specific task. The curve is also highly stable against suite composition changes: iteratively dropping any single task family from the suite shifts Mythos Preview (early)’s expected success share by no more than two points (59 to 61 per cent).

To contextualize this out-of-domain transfer, every headline table isolates an in-domain check restricted to Computer and Mathematical occupations (SOC 15, 5.9 per cent of the wage bill), where the capability assumption holds with the least friction.

4 Results

Table 4 presents the baseline estimates under both frontier models. Moving from Claude Opus 4.6 to Mythos Preview (early) increases the expected-success share of the US wage bill from 52 to 60 per cent. A compound bootstrap procedure resampling the METR curve, adding measurement noise to AEI durations, and re-running the imputation yields a 95 per cent interval of 52 to 66 per cent under the latest frontier.

The share of the wage bill in tasks strictly shorter than the 80 per cent horizon exhibits extreme volatility, jumping from 9 to 31 per cent as the frontier advances. Conversely, the 50 per cent threshold share remains mechanically saturated, encompassing virtually the entire non-physical wage bill.

Table 4: Headline numbers, May 2026. “Non-physical” = no physical work activity. Durations imputed where unobserved. With w_N the non-physical share of the wage bill and $\bar{P}_N$, $\bar{P}_P$ the wage-weighted mean success probability over non-physical and physical tasks: row 1 = $w_N \bar{P}_N$; row 2 = $\bar{P}_N$; row 3 = $w_N \bar{P}_N + (1 - w_N) \bar{P}_P$. SOC 15 rows provide an in-domain benchmark.

Measure (share of wage bill in...)	Denominator	Frontier model	
		Claude Opus 4.6 (Feb 2026; p50 12.0 h, p80 1.2 h)	Mythos Preview (early) (Apr 2026; p50 17.4 h, p80 3.1 h)
Expected success	all tasks	52%	60%
Expected success	non-physical tasks	67%	77%
Expected success, physical tasks scored by the curve rather than excluded	all tasks	70%	79%
Shorter than 80% horizon	all tasks	9%	31%
Shorter than 80% horizon	non-physical tasks	11%	40%
Shorter than 50% horizon	all tasks	75%	78%
Shorter than 50% horizon	non-physical tasks	96%	100%
Observed in Claude usage, May 2026	all tasks	25%	25%
Observed in any AEI release, Dec 2024 to May 2026	all tasks	44%	44%
Expected success, in-domain check	SOC 15 tasks	60%	69%
Shorter than 80% horizon, in-domain check	SOC 15 tasks	3%	17%
Observed in any AEI release, in-domain check	SOC 15 tasks	76%	76%

4.1 The task duration distribution

Figure 2 illustrates the mechanics behind the threshold dynamics. The vast majority of the wage bill falls between 15 minutes and 16 hours. Because the 50 per cent horizons of both models sit to the right of this mass, the p_{50} threshold captures nearly all non-physical tasks.

This near-saturation is an artifact of the AEI duration data, not an absolute capability ceiling. AEI durations measure modular pieces of work delegated in chat interfaces, which are short by selection. Being strictly shorter than the p_{50} horizon only implies that the fitted curve provides better-than-even odds. Indeed, these same non-physical tasks yield a wage-weighted expected success of 67 per cent under Claude Opus 4.6 and 77 per cent under Mythos Preview (early), rather than 100 per cent.

Table 5: Compound 95 per cent intervals from 500 nested bootstrap draws, accounting for METR curve fit uncertainty, AEI duration measurement noise, and imputation variance.

Measure	Error source resampled	Claude Opus 4.6 point (95% interval)	Mythos Preview (early) point (95% interval)
Expected success, all tasks	METR curve (family, task, run)	52% (45 to 60)	60% (53 to 67)
	Durations (measurement noise and imputation)	52% (51 to 52)	60% (59 to 60)
	Both	52% (44 to 60)	60% (52 to 66)
Shorter than 80% horizon, all tasks	METR curve (family, task, run)	9% (1 to 25)	31% (13 to 60)
	Durations (measurement noise and imputation)	9% (7 to 11)	31% (28 to 34)
	Both	9% (1 to 26)	31% (14 to 59)
Expected success, non-physical tasks	Both	67% (56 to 76)	77% (66 to 85)

Because METR explicitly flags horizons beyond 16 hours as unreliable, the integrated expected-success share provides the most stable macroeconomic aggregate. The p_{80} share remains highly sensitive to the model’s confidence interval precisely because its horizon intersects the densest region of the duration distribution (Appendix E).

4.2 Capability over time

Figure 3 traces the capability metrics as the frontier advanced from 2023 to 2026. The expected-success share grew steadily from 17 per cent under o1 to 60 per cent under Mythos Preview (early). The sharp discrete jump in the p_{80} threshold share during the final step is driven by three compounding factors detailed in Appendix E: the frontier is strictly defined by the p_{50} horizon, Claude Opus 4.6 possesses an unusually flat success curve, and the p_{80} horizon crossed the mode of the task duration distribution during this specific transition.

4.3 Disentangling capability and observed use

Figure 4(a) contrasts theoretical capability with actual workplace adoption across each AEI data window. While observed usage within any single snapshot window plateaued between 21 and 27 per cent of the wage bill, the cumulative union across all releases has expanded to 44 per cent. The cumulative measure is highly sensitive to the usage definition: requiring a task to reach one in ten thousand conversations cuts the union to 20 per cent, while adding API traffic raises it to 49 per cent (Appendix Table 15).

Figure 4(b) cross-tabulates the May 2026 wage bill. Crucially, observed use is not a strict subset of capability. At the 80 per cent horizon of Mythos Preview (early), 26 per cent of the wage bill sits beyond the capability frontier but still appears in Claude usage, indicating that workers use AI to augment pieces of complex tasks the agent cannot complete autonomously. Conversely, 16 per cent of the wage bill falls inside the capability frontier but remains entirely untouched by users.

This gap cannot be interpreted as a standard technology adoption lag. The usage metric encompasses out-of-frontier augmentation while excluding untapped within-frontier capability. Furthermore, AEI data is extensive-margin only: it records whether a task appears in a chat log, not the intensity or labor-hour volume of the AI’s contribution.

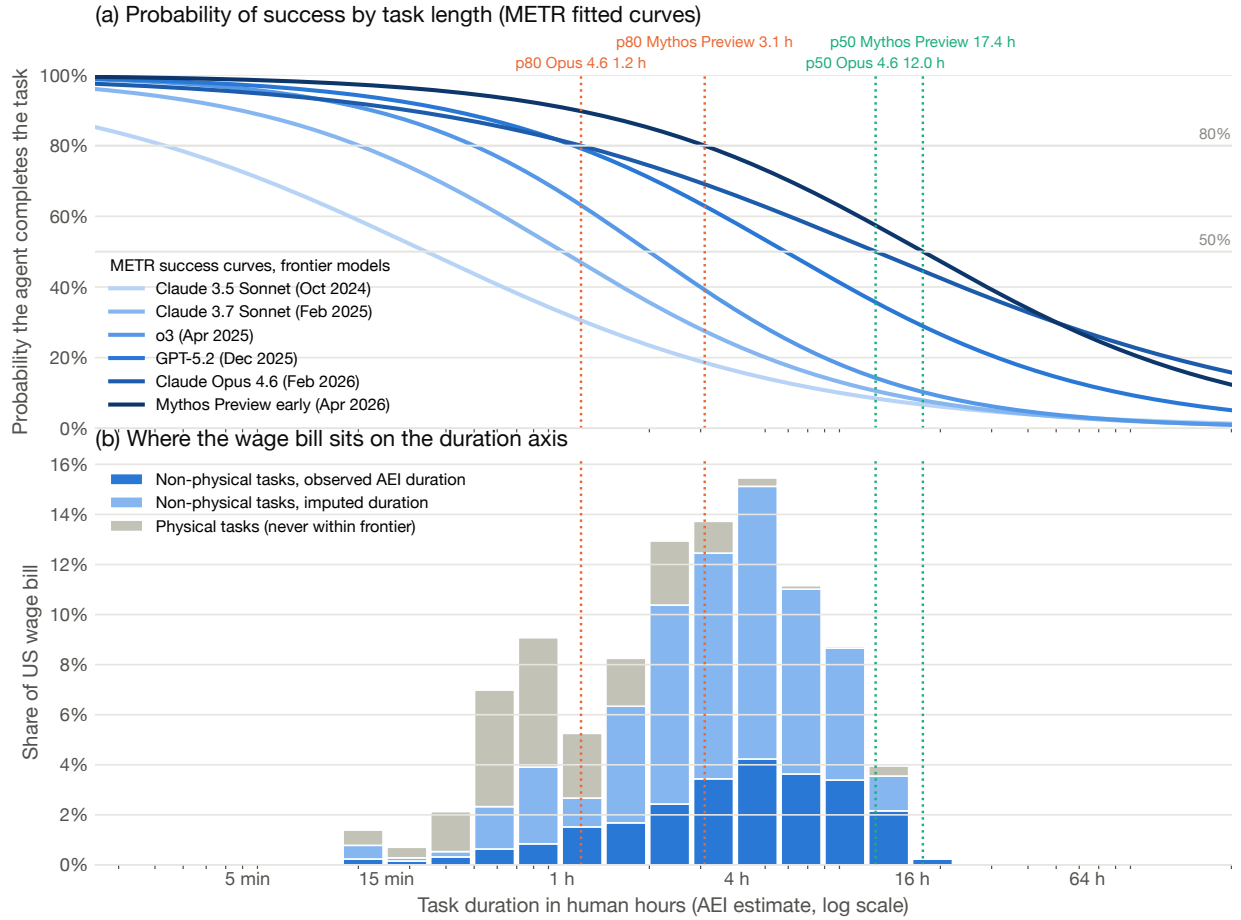


Figure 2: (a) METR’s fitted success curves for six frontier models. (b) The US wage bill by task duration on the same axis, in half-log₂ bins. Dotted lines: the 50 and 80 per cent horizons. The two 80 per cent horizons bracket 22 per cent of the wage bill.

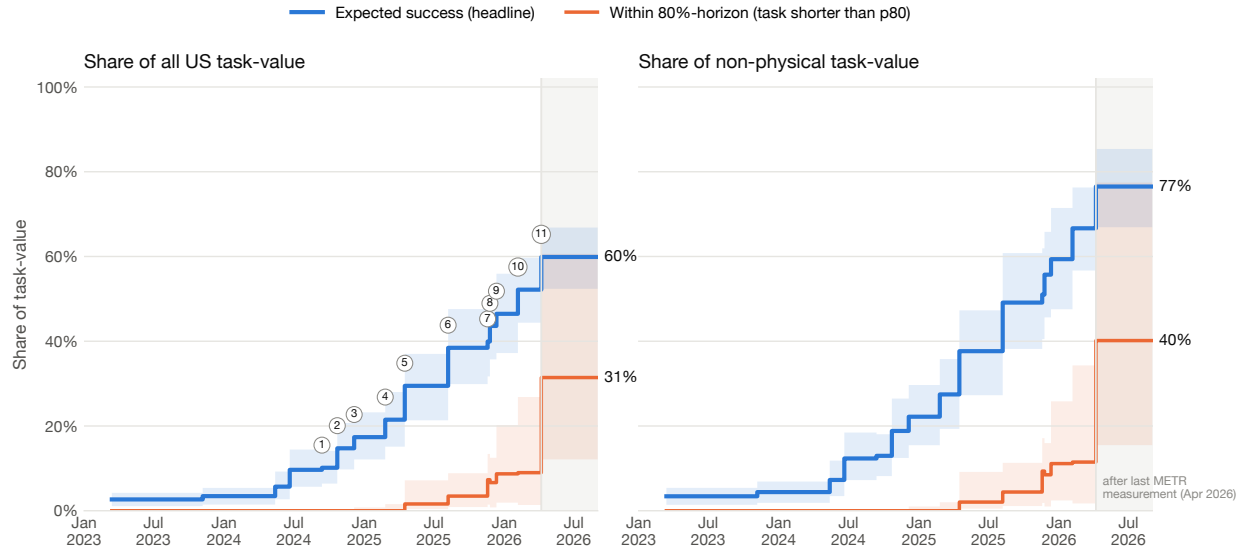


Figure 3: Share of the wage bill in tasks the frontier agent is expected to complete (blue) and in tasks shorter than the 80 per cent horizon (orange). Left: denominator is all tasks. Right: denominator is non-physical tasks only. Frontier = longest 50 per cent horizon among models released by each date. Bands = 95 per cent intervals from the compound bootstrap.

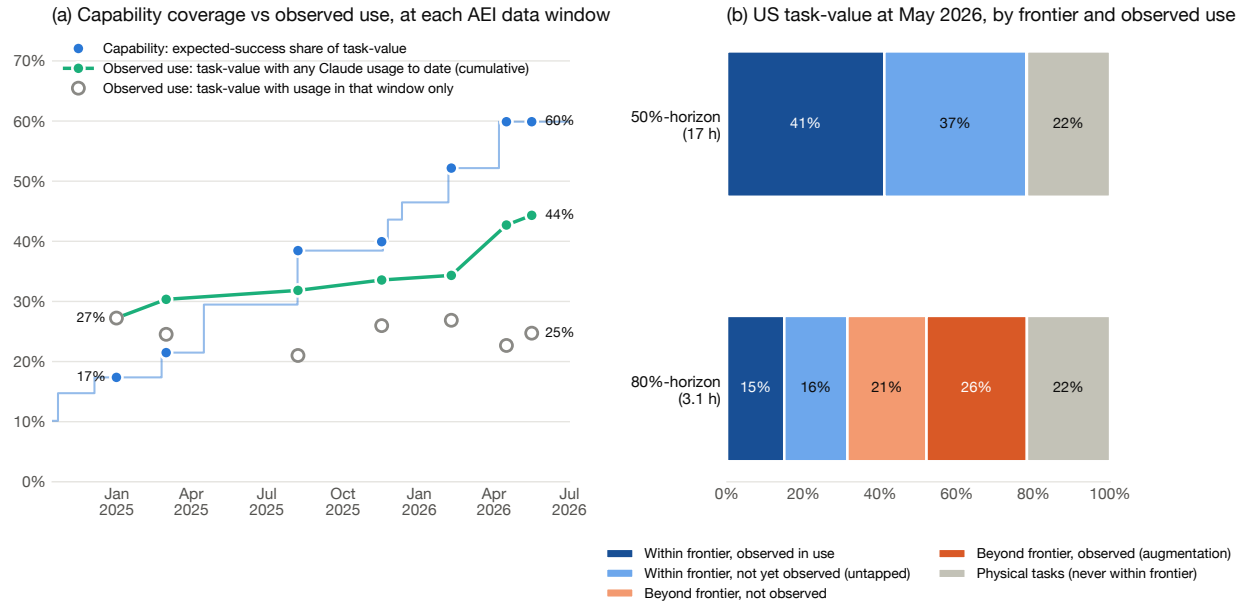


Figure 4: (a) At each AEI data window: expected-success share of the wage bill under the frontier as of the window's last day (blue); share of the wage bill in tasks with a published Claude.ai usage share in any window to date (green) and in that window only (hollow grey). (b) The May 2026 wage bill split by whether a task is shorter than the horizon and whether it has appeared in any AEI release, for the 50 and 80 per cent horizons of Mythos Preview (early).

4.4 Heterogeneity by occupation group

Figure 5 visualizes these measures across the 22 SOC major groups. Sales, legal, office, business, and computer occupations exhibit the highest expected-success shares and the highest observed usage. Construction, cleaning, food service, and installation occupations rank lowest, mechanically driven by the strict physical filter. Within the in-domain Computer and Mathematical occupations, the expected-success share reaches 69 per cent, with 76 per cent of the group’s wage bill appearing in cumulative Claude usage.

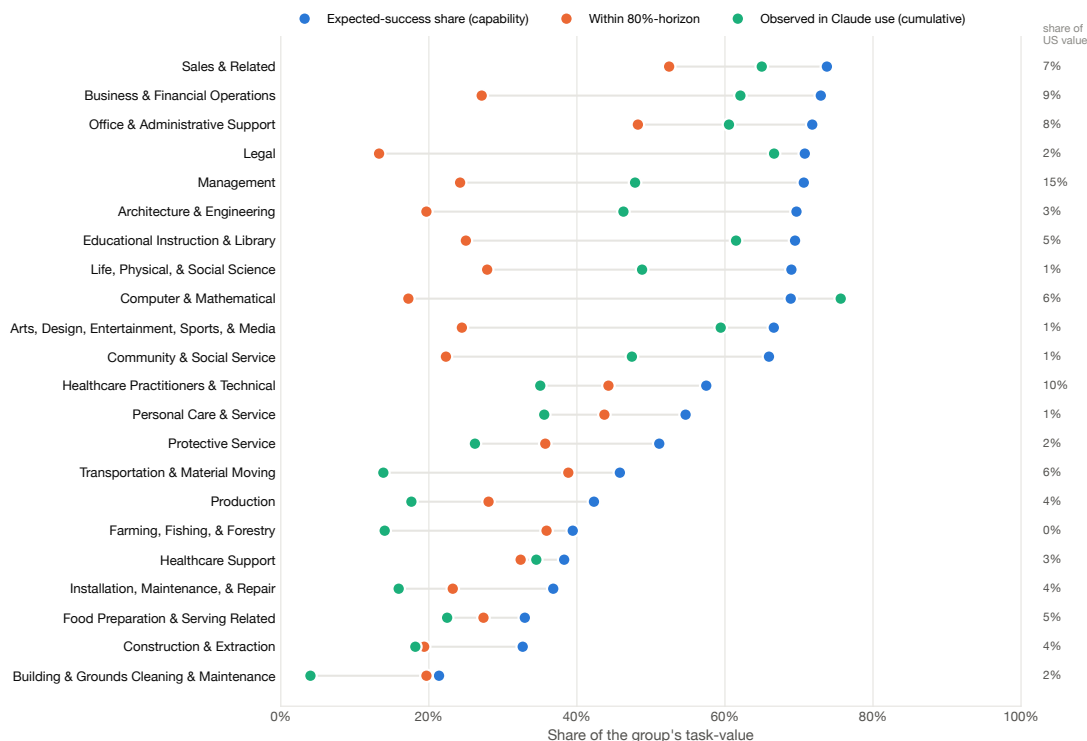


Figure 5: By SOC major group, under Mythos Preview (early): expected-success share (blue), share in tasks shorter than the 80 per cent horizon (orange), share with a published usage share in any AEI release (green). Denominator: the group’s own wage bill, physical tasks included.

Appendix F varies the underlying structural assumptions systematically. Across all variants maintaining the baseline denominator, the expected-success share ranges from 49 to 66 per cent, with the physical exclusion rule and the duration scaling producing the largest shifts.

5 Discussion

5.1 From tasks to jobs

Because an occupation is a bundled set of tasks, task-level coverage metrics do not translate directly into occupational automation. To translate capability into occupational exposure, I compute each 6-digit occupation’s expected-success share and determine the proportion of US employment sitting above specific thresholds (Table 16). Under Mythos Preview (early), 64 per cent of US employment resides in occupations where at least half of the task value is expected to succeed, and 12 per cent

where at least three quarters succeeds. Figure 6 illustrates this distribution for the 80 per cent threshold.

Crucially, no occupation reaches the 90 per cent threshold under either model. On the non-physical denominator, the maximum occupational exposure reaches 97 per cent, demonstrating that the logistic success curve itself binds the ceiling: it remains below unity at every duration, meaning an occupation would require nearly all of its value to reside in tasks of a few minutes to clear 90 per cent. Furthermore, nearly all occupations retain some physical, spatial, or in-person requirements that enforce the zero-probability physical filter.

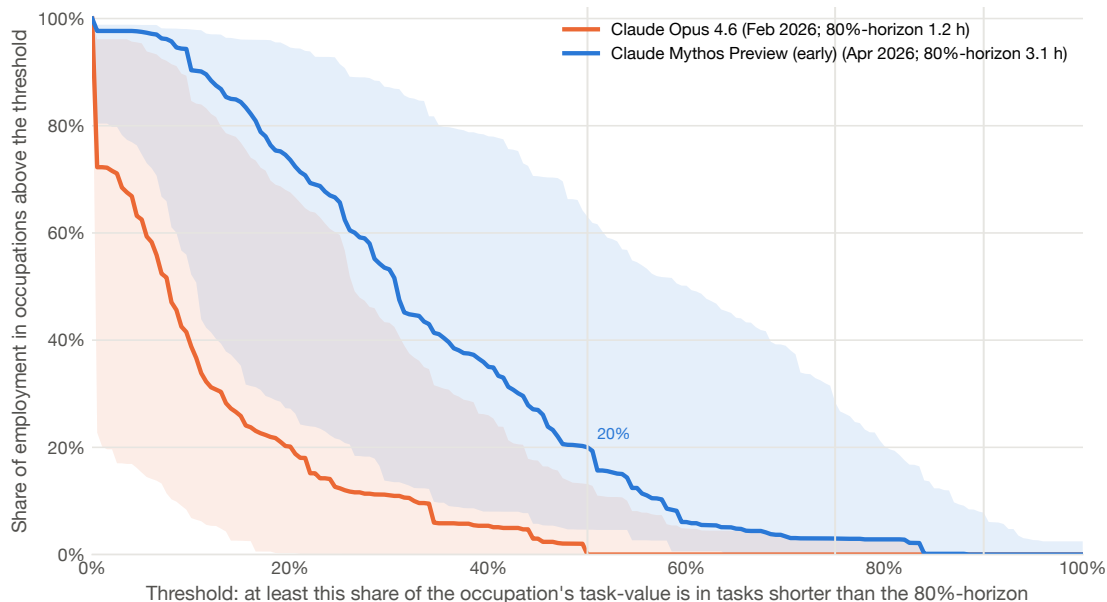


Figure 6: Share of employment in 6-digit occupations where at least the threshold share of the occupation’s task value is in tasks shorter than the frontier model’s 80 per cent horizon. Denominator: 153.4 million jobs. Bands: 95 per cent bootstrap intervals from the frontier model’s fitted curve.

The additive aggregate provides an accurate measure of displaced labor only if tasks can be cleanly unbundled and peeled off a job one at a time. If the uncovered tasks constitute the core value of the job, high task-level coverage may change occupational demand very little until the entire bundle is covered (Autor, 2015). Recomputing the distribution using a weakest-link aggregator (the lowest success probability among an occupation’s core tasks) drastically reduces exposure, placing only 30 per cent of employment above the 50 per cent threshold.

Therefore, an expected-success share of 60 per cent implies partial capability across most occupations, rather than the complete displacement of a few. Determining whether this partial coverage resolves into labor substitution or augmentation requires firm-level reorganization data beyond the scope of this paper.

5.2 Comparison with exposure measures

Across 782 occupations, the rank correlation between the expected-success share and the value-weighted Eloundou et al. (2023) exposure score is 0.79. Much of this agreement is driven by the physical filter; conditional on being non-physical, the rank correlation between expected-success

Table 6: The ten occupations with the highest expected-success share of task value under Mythos Preview (early), and the five largest occupations by employment. Weakest link: lowest success probability among the occupation’s above-median-importance tasks, zero if any is physical. Median duration: wage-weighted baseline durations.

SOC	Occupation	Employment (thousands)	Expected success	Weakest link	Observed use	Median duration (h)
39-3091	Amusement and Recreation Attendants	398	85%	75%	29%	1.8
43-3011	Bill and Account Collectors	159	85%	72%	63%	1.7
45-4023	Log Graders and Scalers	3	85%	79%	0%	1.4
43-4151	Order Clerks	75	85%	74%	45%	1.7
43-4021	Correspondence Clerks	4	85%	74%	72%	1.7
45-2011	Agricultural Inspectors	14	84%	83%	51%	2.4
43-5061	Production, Planning, and Expediting Clerks	390	84%	78%	44%	1.7
43-5032	Dispatchers, Except Police, Fire, and Ambulance	203	84%	73%	39%	2.1
41-9041	Telemarketers	58	84%	80%	75%	2.7
13-2081	Tax Examiners and Collectors, and Revenue Agents	57	83%	86%	33%	1.8
31-1120	Home Health and Personal Care Aides	4,306	39%	0%	50%	1.5
41-2031	Retail Salespersons	3,898	60%	0%	58%	2.1
35-3023	Fast Food and Counter Workers	3,854	29%	0%	16%	0.6
11-1021	General and Operations Managers	3,503	69%	62%	34%	7.0
29-1141	Registered Nurses	3,380	64%	0%	39%	2.9

and the Eloundou score is -0.48. The time-horizon axis therefore adds task duration to the exposure literature, systematically demoting occupations whose non-physical tasks are long (e.g., software developers, median task duration 8.1 hours) while promoting clerical and front-line service occupations whose tasks are short. Figure 10 in Appendix G plots each of these relationships.

5.3 Structural Limitations

These estimates are constrained by the underlying measurement infrastructure. Foremost is the duration mismatch: METR measures human time for complete, fully specified tasks performed by contractors, whereas AEI estimates time for the specific pieces of work users delegate in chat interfaces. Because delegated pieces are short by selection, AEI durations may systematically understate employer-defined task lengths, biasing the capability shares upward. Doubling every duration lowers the expected-success share under Mythos Preview (early) to 52 per cent (Appendix F). Calibrating this bridge directly—by running AEI’s duration estimator on METR’s task prompts—is necessary to determine the exact elasticity of the bias. Furthermore, only 26 per cent of the wage bill has an empirically observed AEI duration. While hold-out checks confirm the imputation is unbiased within the observed sample (Appendix Table 14), it is unknown if these durations seamlessly transfer to universally un-delegated tasks.

Second, extrapolating METR’s capability curves outside computational work carries inherent risks. Within the suite, the curve overstates success in the critical one-to-four-hour band; reweighting the residuals suggests this artifact alone inflates the headline share by roughly 2 points (Appendix Table 9). Outside the suite, there is no empirical guarantee that success probabilities decay with length at identical rates across disparate sectors. Finally, the upper bound of the frontier remains

poorly measured, introducing massive variance to the 80 per cent threshold share (Appendix E).

Despite these limitations, integrating laboratory success curves with logged usage data onto the US wage bill provides an empirical baseline for macroeconomic scenario modeling. In expectation, the capability frontier covers a majority of the wage bill, while observed usage covers less than half. The resulting labor market exposure is broad but partial, with the frontier encroaching on fractions of most jobs while fully enveloping none. Refining these estimates relies heavily on resolving the duration bridge, requiring human completion times for tasks defined structurally at the occupational level, alongside AI success curves measured strictly outside the software domain.

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A Data and Pipeline Construction

Everything in this paper is produced by one pipeline from the raw downloads (`run_all.sh` in the `projects/value_horizon` directory of the author’s `job-transition` repository); every number in the text is written by the pipeline into a macro file, and every numbered assumption is listed in `ASSUMPTIONS.md`. The pipeline also writes a task-level file, `output/tables/task_level_all.csv`, with one row per O*NET task carrying the wage weight, the observed and baseline duration and its source, the physical flags, the Eloundou label, the fitted success probability under both frontier models and the usage flags, so that every aggregate here can be recomputed under other choices.

- **METR:** metr.org/time-horizons, files `benchmark_results_1_1.yaml` and `task_results_1_1.yaml` (updated 8 May 2026), plus github.com/METR/eval-analysis-public for the method (`reports/time-horizon-1-1`). Refit: weighted logistic regression of run success on $\log_2$ human minutes, weights = family weight / runs per task normalised to one per model, regularisation 10^{-5} ; 500 bootstrap draws over family, task and run. Frontier at date d = longest 50 per cent horizon among models released on or before d ; the frontier series steps on release dates, and an AEI data window is paired with the frontier as of its last day. Curve variants (Appendix F) are refit on the same runs: probit link; logistic with the asymptote fixed at 0.85, 0.90 or 0.95; HCAST tasks only; each of the 79 families dropped in turn.
- **AEI:** huggingface.co/datasets/Anthropic/EconomicIndex. Usage shares by O*NET task from seven Claude.ai data windows (Dec 2024–Jan 2025, Feb–Mar 2025, Aug 2025, Nov 2025, Feb 2026, Apr 2026, May 2026). Per-task human time (`human_only_time_mean`: mean estimated

hours for a human to complete the delegated work without AI, averaged over the conversations mapped to the task) from the June 2026 release (April and May 2026, simple mean of the two monthly means; baseline) and from the Nov 2025 and Feb 2026 releases (pooled; alternative). Releases before June 2026 identify tasks by text; I match text to O*NET task IDs exactly, recovering 87 to 91 per cent of usage, and drop the rest (mostly tasks of occupations since retired from O*NET). The June 2026 release carries task IDs and joins fully. Imputation: for an unobserved task, the simple mean over its detailed work activities of the geometric mean of observed durations in each activity (36 per cent of the wage bill); if no activity has an observed peer, the same at the intermediate (22 per cent) and then the generalized work activity (15 per cent); otherwise the global geometric mean (0.8 per cent). Tasks map to 1.27 detailed work activities on average.

- **O*NET 30.2:** task statements, task ratings (importance, relevance, frequency), tasks to detailed work activities, work-activity hierarchy. Weights use core tasks with positive relevance; specialties are averaged into the 6-digit SOC code; 29 occupations without ratings get equal weights. Physical work activities: 4.A.3.a.1–4 (performing general physical activities, handling and moving objects, controlling machines and processes, operating vehicles and equipment), 4.A.3.b.4–5 (repairing and maintaining equipment) and 4.A.4.a.5 (assisting and caring for others).
- **OEWS May 2025, national:** employment and mean annual wage by detailed occupation (hourly wages annualised by OEWS at 2,080 hours); 12 OEWS hybrid codes mapped explicitly. Occupations without O*NET tasks hold 1.3 per cent of employment and are dropped.
- **Eloundou et al. (2023) labels:** github.com/openai/GPTs-are-GPTs, used for the alternative definition of the excluded set and for the comparison in Section 5.2.

B METR Task Suite and Refit Verification

Table 7 gives the composition of METR’s Time Horizon 1.1 suite by source. Table 8 reports METR’s published 50 and 80 per cent horizons alongside the refit, and the fitted slope per doubling of task length, for all models with fitted curves. “Frontier at release” is METR’s `is_sota` indicator; these models are the frontier steps in Figure 3. Table 9 reports the fit’s residuals by duration bin.

Table 7: Composition of METR’s Time Horizon 1.1 task suite. Weight = share of METR’s diversity-adjusted weight in the Mythos Preview (early) fit, where each task in a family of n_f tasks carries weight $1/\sqrt{n_f}$, so a family’s total weight is proportional to $\sqrt{n_f}$. The band from half an hour to eight hours holds 40 per cent of the weight on 72 tasks.

Source	Content	Tasks	Families	Human time: median, max	Weight
HCAST	software engineering, machine learning, cybersecurity, and computer-based research tasks	157	69	44 min, 30.0 h	81%
RE-Bench	machine-learning research engineering tasks	5	5	8.0 h, 8.2 h	4%
SWAA	seconds-long software actions (code completion, file selection, request routing)	66	5	5 s, 43 s	15%

Table 8: METR’s published 50 and 80 per cent horizons and my refit from the per-task results, all models with fitted curves. Slope = the refit coefficient on $\log_2$ human minutes, sign reversed: the fall in the log-odds of success per doubling of task length. “Frontier at release” is METR’s `is_sota` flag; these models are the numbered steps in Figure 3, in date order from GPT-4 (March 2023). Mythos Preview (early) is a pre-release checkpoint measured on 1,149 runs.

Model	Released	p50 METR (min)	p50 refit (min)	p80 METR (min)	p80 refit (min)	Slope	Frontier at release
GPT-4 0314	2023-03-14	4.0	4.0	0.9	0.9	0.64	yes
GPT-4 1106	2023-11-06	4.0	4.0	0.8	0.8	0.59	yes
Claude 3 Opus	2024-03-04	4.0	4.0	0.6	0.6	0.53	
GPT-4 Turbo	2024-04-09	3.7	3.7	0.9	0.9	0.69	
GPT-4o	2024-05-13	7.0	7.0	1.3	1.3	0.56	yes
Claude 3.5 Sonnet (Old)	2024-06-20	11.4	11.4	1.7	1.7	0.50	yes
o1-preview	2024-09-12	20.3	20.3	4.4	4.4	0.63	yes
Claude 3.5 Sonnet (New)	2024-10-22	20.5	20.5	2.6	2.6	0.46	yes
o1	2024-12-05	38.8	38.8	7.1	7.1	0.57	yes
Claude 3.7 Sonnet	2025-02-24	60.4	60.4	12.1	12.1	0.60	yes
o3	2025-04-16	119.7	119.7	30.0	30.0	0.69	yes
Claude 4 Opus	2025-05-22	100.4	100.4	20.4	20.4	0.60	
Claude 4.1 Opus	2025-08-05	100.5	100.5	23.5	23.5	0.66	
GPT-5	2025-08-07	203.0	203.0	38.3	38.3	0.58	yes
Gemini 3 Pro	2025-11-18	224.3	224.3	54.1	54.1	0.68	yes
GPT-5.1-Codex-Max	2025-11-19	223.7	223.7	50.6	50.6	0.65	
Claude Opus 4.5	2025-11-24	293.0	293.0	49.4	49.3	0.54	yes
GPT-5.2	2025-12-11	352.2	352.3	66.0	66.0	0.57	yes
GPT-5.3-Codex	2026-02-05	349.5	349.5	54.7	54.7	0.52	
Claude Opus 4.6	2026-02-05	718.8	718.8	69.9	69.9	0.41	yes
Gemini 3.1 Pro	2026-02-19	384.1	384.1	89.8	89.8	0.66	
GPT-5.4	2026-03-05	341.7	341.7	53.9	53.9	0.52	
Claude Mythos Preview (early)	2026-04-07	1044.8	1044.7	185.9	185.9	0.56	yes

Table 9: Observed minus fitted success by duration bin within METR’s suite, Mythos Preview (early). Observed and fitted are diversity-weighted means over the tasks in the bin; difference in percentage points. Last column: share of the total wage bill in non-physical tasks whose AEI duration falls in the bin. Reweighting the differences by the last column gives the change in the expected-success share that would follow if the suite’s residual pattern held in the economy: about -2 points for Mythos Preview (early) and -1 for Claude Opus 4.6.

Human time	METR suite		Mythos Preview (early)			Non-physical wage bill in this bin
	Tasks	Weight	Observed success	Fitted	Difference	
≤ 1 min	67	15%	99%	100%	-0.5	0%
1-15 min	52	24%	99%	98%	+0.3	1%
15-60 min	36	17%	98%	94%	+3.3	7%
1-4 h	18	9%	76%	86%	-9.9	32%
4-16 h	50	31%	68%	67%	+1.3	38%
≥ 16 h	5	3%	45%	47%	-1.4	0%

C Task Examples and Classification

Table 10 lists the twelve METR task families with the largest weight in the fit; the family names are METR’s identifiers, and Rein et al. (2025) and Wijk et al. (2024) describe the tasks. Table 11 shows, for eight SOC major groups, the O*NET task with the largest wage bill in the group, with its duration on the baseline axis. Figure 7 shows every O*NET task of one occupation, Registered Nurses, with the physical filter applied.



Figure 7: The 27 O*NET tasks of Registered Nurses (O*NET-SOC 29-1141.00; nursing specialties excluded here). Each tile is one task statement, shortened. Dark blue: no physical work activity. Grey with orange border: some physical work activities, so counted as physical. Grey: every work activity physical. Physical tasks never count as within the frontier. Grey text: the task's AEI human duration on the baseline axis, the estimated human time for the work delegated in a conversation, averaged over conversations; * = imputed. Tiles are ordered by classification and then by O*NET importance.

Table 10: The twelve METR task families with the largest diversity-adjusted weight in the Mythos Preview (early) fit.

Task family	Source	Tasks	Human time, shortest to longest	Weight
<code>arithmetic</code>	SWAA	20	1 s to 43 s	3.7%
<code>file_selection</code>	SWAA	20	2 s to 6 s	3.7%
<code>alert_triage</code>	SWAA	10	4 s to 10 s	2.6%
<code>env_scientist</code>	HCAST	9	44 min to 11.1 h	2.5%
<code>request_routing</code>	SWAA	9	8 s to 18 s	2.5%
<code>pico.ctf</code>	HCAST	7	2 min to 1.1 h	2.2%
<code>data_cleaning_arjun</code>	HCAST	7	6 min to 6 min	2.2%
<code>code_completion</code>	SWAA	7	3 s to 6 s	2.2%
<code>palisade_crackme</code>	HCAST	6	21 min to 8.0 h	2.0%
<code>blackbox</code>	HCAST	6	10 min to 1.1 h	2.0%
<code>password_check</code>	HCAST	6	57 s to 32 min	2.0%
<code>hypothesis_testing</code>	HCAST	6	9 min to 49 min	2.0%

Table 11: The O*NET task with the largest wage bill in each of eight SOC major groups. Hours = AEI human duration on the baseline axis: the estimated time for a person to do, without AI, the piece of the task that users delegate in one conversation, averaged over conversations mapped to the task; * = imputed. The wage weight attaches to the whole statement. Physical = at least one of the task’s work activities is physical, so the task never counts as within the frontier.

Occupation	Task (O*NET statement)	Hours	Physical
General and Operations Managers	Review financial statements, sales or activity reports, or other performance data to measure productivity or goal. . .	7.0	
Software Developers	Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.	11.3	
Lawyers	Interpret laws, rulings and regulations for individuals and businesses.	4.3	
Registered Nurses	Record patients’ medical information and vital signs.	2.0*	
Retail Salespersons	Greet customers and ascertain what each customer wants or needs.	0.5	yes
Customer Service Representatives	Confer with customers by telephone or in person to provide information about products or services, take or enter orders. . .	0.8	
Construction Laborers	Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials. . .	2.1*	yes
Laborers and Freight, Stock, and Material Movers, Hand	Maintain equipment storage areas to ensure that inventory is protected.	0.6*	

D Variation within METR’s Suite

Figure 8 plots, for the two frontier models, each METR task’s observed success rate against its human time, with METR’s pooled curve and curves fitted within each task source. Table 12 gives the horizons from each fit and the range of horizons obtained by dropping any one task family and refitting. The R^2 in the panel titles is the weighted R^2 of task success rates on the pooled curve; the corresponding ceiling if the curve were exactly right, given the number of runs per task, is 0.61 for Claude Opus 4.6 and 0.57 for Mythos Preview (early). Across all 23 models with fitted curves the highest weighted R^2 is 0.68 (GPT-4 0314, ceiling 0.89); the frontier models sit low because most of their tasks are solved nearly every time and the remaining failures are concentrated in particular tasks rather than at particular lengths. The RE-Bench fits are not shown because five tasks do not identify a slope. The SWAA-only horizons are shown but should not be read as measurements: the longest SWAA task takes 43 seconds, so both horizons for Mythos Preview (early) and the 50 per cent horizon for Claude Opus 4.6 lie beyond the subsuite’s data, and one percentage point of observed success separates a one-minute from a sixteen-minute fitted horizon.

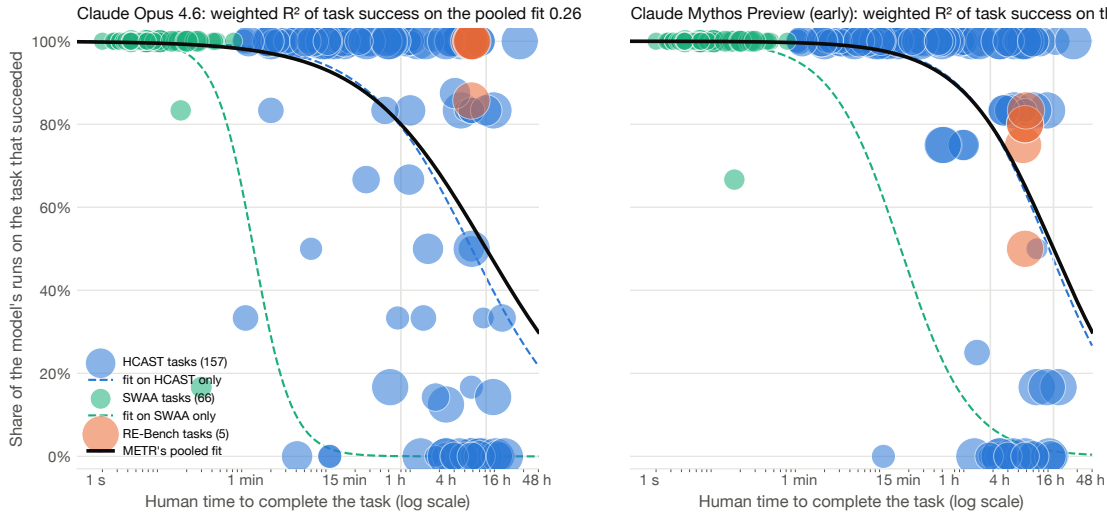


Figure 8: Each point is one METR task; area is proportional to METR’s diversity weight. Black: METR’s pooled fit. Dashed: logistic fits within one task source. Grey lines: the pooled 50 and 80 per cent horizons. Left: Claude Opus 4.6. Right: Mythos Preview (early).

E Decomposing the April 2026 Capability Step

The 80 per cent threshold share in Figure 3 jumps from 9 to 31 per cent when the frontier passes from Claude Opus 4.6 to Mythos Preview (early). The step is a faithful reading of METR’s published fits, but three things make it as large as it is. First, the frontier only moves when a model beats the best 50 per cent horizon to date. METR measured two other models between these two: Gemini 3.1 Pro (19 February 2026; 50 per cent horizon 6.4 hours, 80 per cent horizon 1.5 hours) and GPT-5.4 (5 March 2026; 50 per cent horizon 5.7 hours, 80 per cent horizon 0.9 hours). Neither beat the 50 per cent horizon of Claude Opus 4.6, so neither appears in the figure, although the 80 per cent horizon of Gemini 3.1 Pro was the longer of the two and would have put 11 per cent of the wage bill below it; its expected-success share, 49 per cent, is below Claude Opus 4.6’s, so the frontier rule and the share agree on the ranking. Second, Claude Opus 4.6 has an

Table 12: Horizons from METR’s pooled fit, from fits within one task source, and the range when any one of the 79 task families is dropped. Success = diversity-weighted share of runs that succeeded. †: horizon lies beyond the longest task in the subsuite (an extrapolation, not a reading within the data).

Model	Fit	Tasks	Weight	Success	50% horizon	80% horizon
Claude Opus 4.6	pooled (METR’s fit)	228	100%	79%	12.0 h	1.2 h
	HCAST tasks only	157	81%	74%	8.1 h	1.1 h
	SWAA tasks only	66	15%	98%	1 min [†]	36 s
	RE-Bench tasks only	5	4%	97%	n/a	n/a
	drop any one family				10.3 h to 14.2 h	1.1 h to 1.4 h
Mythos Preview (early)	pooled (METR’s fit)	228	100%	85%	17.4 h	3.1 h
	HCAST tasks only	157	81%	83%	15.4 h	3.1 h
	SWAA tasks only	66	15%	99%	16 min [†]	4 min [†]
	RE-Bench tasks only	5	4%	74%	n/a	n/a
	drop any one family				14.8 h to 20.9 h	2.9 h to 3.6 h

unusually flat success curve, 0.41 per doubling against 0.56 for Mythos Preview (early). Its 50 per cent horizon is 10 times its 80 per cent horizon, against 4 to 6 times for the other frontier models since 2025, so its 80 per cent horizon barely moved from that of GPT-5.2 (1.1 to 1.2 hours) while its 50 per cent horizon doubled. Much of the 8-point rise in the expected-success share between the two models is therefore the slope steepening in the half-hour-to-eight-hour band rather than the horizon lengthening. Third, the 80 per cent threshold share counts tasks shorter than a threshold, and the threshold moved from 1.2 to 3.1 hours, across the part of the duration axis where the wage bill is: 22 per cent of the wage bill is in non-physical tasks with durations between those two values (Figure 2(b)), and 32 per cent is in non-physical tasks between one and four hours. The expected-success share integrates the curve instead of counting a threshold, and moves 8 points on the same step.

The threshold is also poorly determined at the top. The 80 per cent horizon of Mythos Preview (early) has a 95 per cent interval of 1.6 to 6.6 hours, which puts the share anywhere between 12 and 61 per cent (the orange band in Figure 3). Under Claude Opus 4.6 the interval is 1 to 27 per cent. The level of the orange line in 2026 is therefore uncertain by tens of points either way, and the size of the step says as much about where the wage bill sits on the duration axis as about the model.

F Robustness and Sensitivity Analysis

Table 13 changes one assumption at a time, with Mythos Preview (early) as the frontier, in five groups. Across the variants that keep the baseline denominator, the expected-success share ranges from 49 to 66 per cent. The duration bridge and the exclusion rule matter most. Doubling every duration gives 52 per cent and cuts the 80 per cent threshold share to 12 per cent; halving them raises that share to 58 per cent. Inflating imputed durations alone, which targets the selection of the observed sample rather than the level of the clock, gives 57, 54 and 51 per cent at factors of 1.5, 2 and 3. Using the earlier AEI releases for durations raises the share by 5 points. On the curve side, refitting on HCAST tasks only, dropping any one family, switching to a probit link or fixing the asymptote at 0.85 to 0.95 all leave the share within two points of the baseline, although the probit and low-asymptote fits move the 80 per cent threshold share to 27 and 23 per cent. Task weights within occupations barely matter, including the hours-implied weight; dropping

wages and weighting by employment gives 55 per cent. The lenient physical filter gives 63 per cent, the Eloundou labels 49 per cent, and crediting mixed tasks’ non-physical share at a proportionally shorter duration 61 to 62 per cent. The two observed-use variants leave the expected-success columns unchanged by construction; what they move is the use column, from 44 to 20 per cent under a share threshold and to 49 per cent with API traffic. The last two rows restrict the universe to tasks with an observed duration and are not commensurable with the rest; they are the source of the 75 per cent that appears if the whole column is read as one range.

Table 13: One assumption changed at a time; frontier = Mythos Preview (early). Denominator: all tasks, except in the last two rows, where it is the observed-duration tasks. Curve variants refit the same model’s runs another way (Appendix A). θ : share of a mixed task’s value credited as non-physical, evaluated at duration θD_t .

Variant (one change from baseline)	Expected success	< p80	Observed, cumulative
<i>Duration bridge</i>			
Baseline	60%	31%	44%
Durations from the Nov 2025 and Feb 2026 AEI releases	64%	51%	44%
All durations halved	66%	58%	44%
All durations doubled	52%	12%	44%
Imputed durations only $\times 1.5$	57%	21%	44%
Imputed durations only $\times 2$	54%	17%	44%
Imputed durations only $\times 3$	51%	14%	44%
<i>Success curve (same model, refit)</i>			
Curve fitted on HCAST tasks only	59%	30%	44%
Drop-one-family refit, lowest share (drops <code>mlab</code>)	59%	29%	44%
Drop-one-family refit, highest share (drops <code>spn_cryptanalysis</code>)	61%	37%	44%
Probit link instead of logistic	59%	27%	44%
Logistic with asymptote $\bar{p} = 0.95$ (refit)	60%	35%	44%
Asymptote $\bar{p} = 0.90$	60%	32%	44%
Asymptote $\bar{p} = 0.85$	59%	23%	44%
<i>Task weights and value</i>			
Equal task weights within occupation	60%	31%	44%
Importance $\times$ relevance weights	60%	32%	45%
Frequency-implied weights	60%	32%	46%
Hours-implied weights (duration $\times$ annual frequency)	60%	21%	46%
Employment weights, no wages	55%	33%	39%
<i>Excluded (physical) set</i>			
Non-physical set = Eloundou et al. exposure label > 0	49%	24%	44%
Physical = every work activity physical (lenient filter)	63%	34%	44%
Mixed tasks split: share $\theta = 0.25$ of value at duration θD	61%	32%	44%
Mixed tasks split: $\theta = 0.5$	61%	33%	44%
Mixed tasks split: $\theta = 0.75$	62%	34%	44%
<i>Observed-use definition (expected success unchanged by construction)</i>			
Observed only if share $\geq 0.01\%$	60%	31%	20%
Observed in Claude.ai or API	60%	31%	49%
<i>Change of universe (different denominator; not part of the one-change range)</i>			
Observed-duration tasks only (denominator changes)	73%	34%	100%
Observed-duration tasks only, Nov 2025 and Feb 2026 AEI releases	75%	56%	100%

Table 14: Leave-one-out check of the duration imputation. Each observed task is re-imputed from the other observed tasks by the baseline rule, at the level the rule would use for it. $\hat{D}_t/D_t$ is the ratio of imputed to observed duration. Value shares are of the wage bill in observed-duration tasks.

Imputation level	Tasks	Share of observed value	Median $\hat{D}_t/D_t$	Mean log ratio, value-weighted	SD of log ratio	Within factor 2
Detailed work activity	2,736	97%	1.00	+0.02	0.49	87%
Intermediate work activity	69	2%	1.57	+0.94	0.78	57%
Generalized work activity	9	0%	2.61	+1.38	0.84	44%
All observed tasks	2,814	100%	1.00	+0.04	0.50	86%

Table 15: Observed use under alternative definitions. Share of the wage bill in tasks meeting the definition; denominator all tasks. Top block: threshold on the task’s share of Claude.ai conversations in a window. Bottom block: number of the seven Claude.ai windows in which the task was published at all.

Definition of “observed”	Tasks	May 2026 window	Any window, Dec 2024 to May 2026
Published at all (baseline)	6,254	25%	44%
Share of conversations $\geq 0.01\%$	2,297	13%	20%
Share $\geq 0.05\%$	700	4%	8%
Share $\geq 0.1\%$	370	2%	4%
Published in at least 1 of the 7 windows	6,254		44%
Published in at least 2 of the 7 windows	5,286		38%
Published in at least 4 of the 7 windows	3,323		24%
Published in at least 7 of the 7 windows	843		9%

G Occupation-Level Exposure and Time Series

Figure 9 plots every 6-digit occupation by its expected-success share of task value under Mythos Preview (early) and the share of its task value that has appeared in Claude usage. Occupations fall into two clusters: office, sales, business and computer work at the top right; service, construction, production and transport work at the bottom left. Large occupations with high capability but use well below the economy-wide share, such as general managers and registered nurses, make up much of the “inside the horizon, not yet observed” cell of Figure 4(b). Table 16 gives the employment shares behind Section 5.1, and Table 17 the capability and use measures at each AEI data window.

Figure 10 plots the correlations of Section 5.2. The agreement in panels (a) to (c) is largely the physical filter, which both measures respect; panels (d) and (e) condition it away by using non-physical value only, leaving a measure that depends on task duration alone. Panel (f) supplies the channel: the occupations the rubric scores as most exposed are the ones whose delegated tasks are longest, and the success curve falls with length. The negative correlation in panel (d) is therefore not disagreement about which occupations are exposed, but the duration axis discounting long analytical work relative to short clerical and service work. Two cautions: the vertical range in panels (d) and (e) is narrow, so the rank correlation turns on small differences and the employment-weighted correlation is much the weaker of the two; and the occupations at the top left of panel (e) are mostly physical occupations whose small non-physical remainder happens to be short.

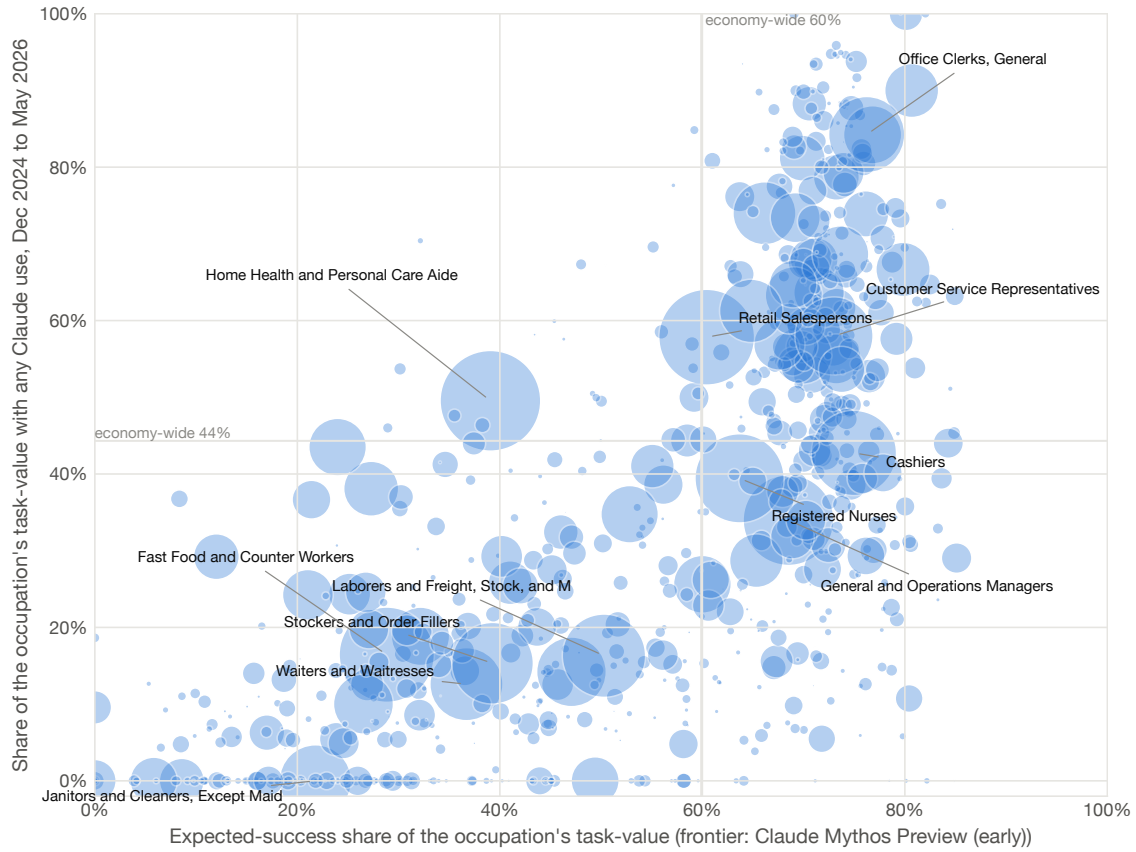


Figure 9: One bubble per 6-digit occupation (782 occupations), area proportional to employment. Horizontal axis: expected-success share of the occupation's task value under Mythos Preview (early). Vertical axis: share of the occupation's task value with a published Claude.ai usage share in any AEI release, December 2024 to May 2026. Grey lines: economy-wide values. Physical tasks count against the occupation on both axes. The 12 largest occupations are labelled.

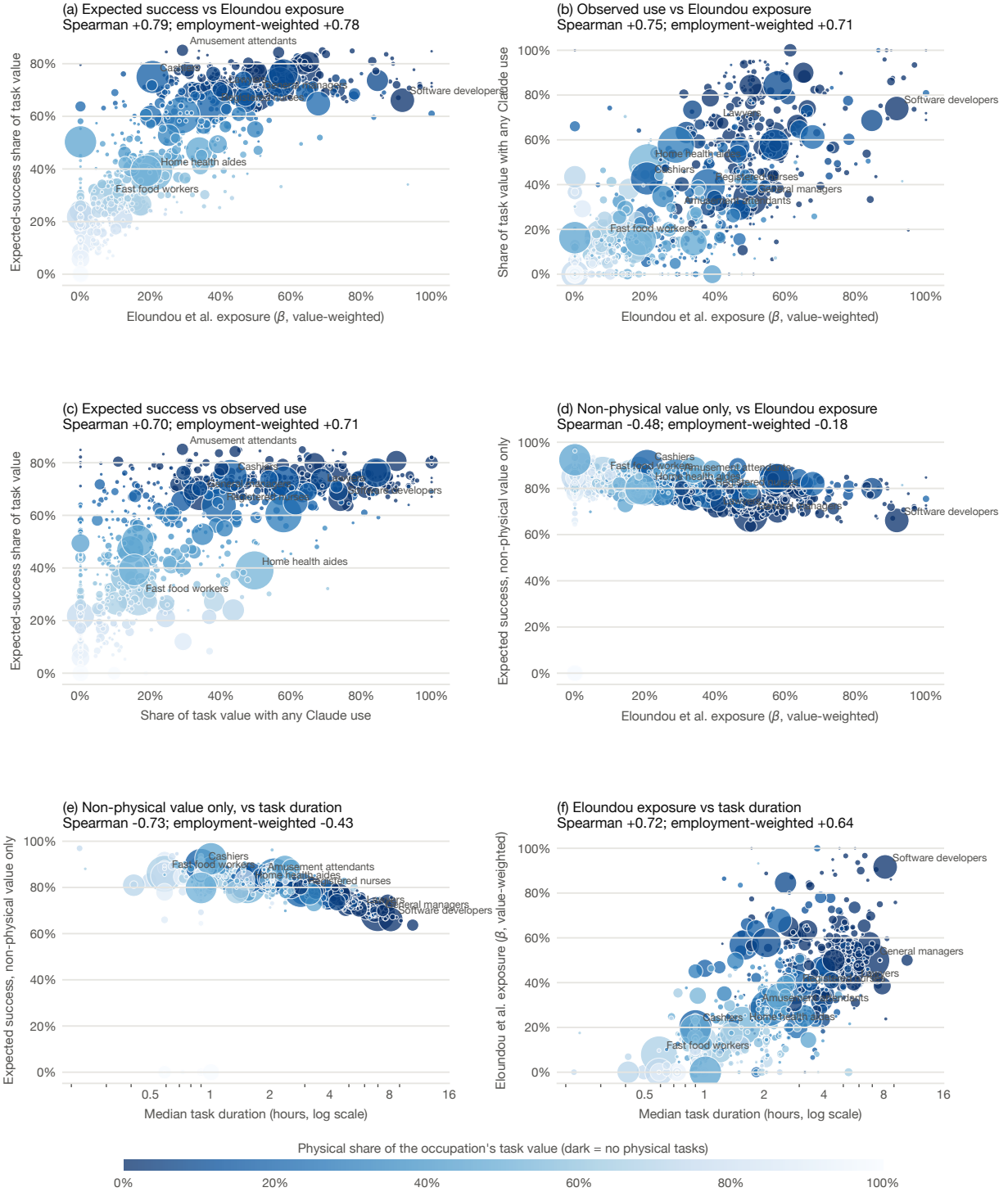


Figure 10: The comparison with rubric-based exposure measures described in Section 5.2. One bubble per 6-digit occupation, area proportional to employment, shade the physical share of the occupation's task value. Each panel prints the unweighted Spearman rank correlation across the 782 occupations and its employment-weighted Pearson counterpart. Panels (a) to (c) use the additive expected-success share, in which physical tasks enter at probability zero. Panels (d) and (e) use the expected-success share of the occupation's non-physical task value only, which is a function of task duration alone. Panel (f) plots exposure against duration. Eloundou exposure is the value-weighted mean of the task-level β labels within the occupation; frontier = Mythos Preview (early).

Table 16: From tasks to jobs. Share of employment (153.4 million jobs) in occupations whose own coverage measure is at least the threshold. In the first three measures and the last two, physical tasks count against the occupation (probability zero); the non-physical measure drops them from the occupation’s denominator. Geometric mean: CES aggregator of task success probabilities with substitution parameter $\rho \rightarrow 0$, weighted by task value. Weakest link: the lowest success probability among the occupation’s tasks with importance at or above its median. Observed use is not conditioned on.

Occupation-level measure	At least	Claude Opus 4.6	Mythos Preview (early)
Expected-success share of the occupation’s task value (additive)	50%	58%	64%
	75%	0%	12%
	90%	0%	0%
Share of the occupation’s task value shorter than the 80% horizon	50%	0%	20%
	75%	0%	3%
	90%	0%	0%
Share of the occupation’s task value shorter than the 50% horizon	50%	70%	70%
	75%	53%	54%
	90%	30%	36%
Expected-success share of the occupation’s <i>non-physical</i> task value	50%	99%	99%
	75%	21%	77%
	90%	0%	3%
Geometric mean of task success (CES, $\rho \rightarrow 0$)	50%	23%	23%
	75%	0%	8%
	90%	0%	0%
Weakest link: lowest success probability among above-median-importance tasks	50%	23%	30%
	75%	0%	2%
	90%	0%	0%

Table 17: Capability and observed use at each AEI data window. Frontier = longest 50 per cent horizon among models released by the window’s last day. Denominator: all tasks. “< p80” = share of the wage bill in non-physical tasks shorter than the frontier model’s 80 per cent horizon.

AEI window	Frontier model	Released	p50 (h)	p80 (h)	Expected success	< p80	Observed, window	Observed, cumulative
Dec 24–Jan 25	o1	5 Dec 2024	0.6	0.1	17%	0%	27%	27%
Feb–Mar 2025	Claude 3.7 Sonnet	24 Feb 2025	1.0	0.2	21%	0%	25%	30%
Aug 2025	GPT-5	7 Aug 2025	3.4	0.6	38%	3%	21%	32%
Nov 2025	Gemini 3 Pro	18 Nov 2025	3.7	0.9	40%	7%	26%	34%
Feb 2026	Claude Opus 4.6	5 Feb 2026	12.0	1.2	52%	9%	27%	34%
Apr 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	60%	31%	23%	43%
May 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	60%	31%	25%	44%